

Le Chen[†] and Nicholas Eisenberg^{*}

Auburn University,

Department of Mathematics and Statistics,

Auburn, Alabama, USA.

Email: le.chen@auburn.edu[†]

nickeisenberg@gmail.com^{*} or nze0019@auburn.edu^{*}

To

Journal of Theoretical Probability,

January 2, 2023

Dear Editor,

In this paper we investigate the conditions required to guarantee the existence of an invariant measure to the stochastic heat equation on the whole space $\mathbb{R}^d$ with an unbounded Lipschitz diffusion term b . To be more precise, we study

$$\partial_t u(t, x) - \frac{1}{2} \Delta u(t, x) = b(u(t, x)) \dot{W}(t, x),$$

with the initial condition $u(0, \cdot) = \mu(\cdot)$, where $\dot{W}$ is the centered Gaussian noise that is white in time and colored in space. The major challenge to the problem is finding conditions which guarantee that the probability moments of the solution in a weighted $L^2(\mathbb{R}^d)$ space are bounded in time, which is often the case when some additional drift terms to the equation are imposed to help control the usual exponential growth of the moments in time [6, 7, 8]. We do not have such a drift term and the literature for such a set up is far more scarce than when considering the added drift term. An immediate consequence of no drift term in our equation is that the spatial dimension d has to be at least 3, i.e., $d \geq 3$.

Our paper is mostly inspired by Tessitore and Zabczyk's work [1], where they prove existence of an invariant measure under the same setting as ours, i.e., on the whole space and with an unbounded and Lipschitz diffusion term. *The first contribution* of our paper is more general initial conditions, including the important Dirac delta initial data. Note that in [1], only the constant initial condition is considered, even though their techniques essentially allow for all bounded functions. We emphasize that the Dirac delta measure plays an important role in the asymptotic analysis for stochastic heat equation; see, e.g., the work of [5]. *The second contribution* is clean conditions on the spectral density of the noise. Indeed, the conditions on the spectral density in [1] are not easy to verify in practice. We identify the following clean and near optimal condition:

$$\int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^2 \wedge |\xi|^{2(1-\alpha)}} < \infty, \quad \text{for some } \alpha \in (0, 1),$$

where f is the correlation function of the noise and $\widehat{f}$ is its Fourier transform — the so-called spectral density; see Remark 2 of the manuscript for more details.

Our improvements are due to the fact that we work under the random field framework of Walsh [3], while in the past, this problem has usually been studied under the theory of the stochastic differential equation in the infinite-dimensional setting laid out by Da Prato

and Zabczyzk [4]. While both theories have been shown to be equivalent [2], they both offer their own unique set of tools to work with. Especially the random field framework provides sharper moment estimates on the moments of the solution, which is crucial to our analysis.

We hope that the content and contributions of the paper fit well the journal and we look forward to hearing from you. Thank you very much for handling our paper.

Sincerely,

Le Chen and Nicholas Eisenberg

References

- [1] Tessitore, G., Zabczyzk, J.: Invariant measures for stochastic heat equations. *Probab. Math. Statist.* **18**(2, Acta Univ. Wratislav. No. 2111), 271–287 (1998)
- [2] Dalang, R.C., Quer-Sardanyons, L.: Stochastic integrals for spde’s: a comparison. *Expo. Math.* **29**(1), 67–109 (2011). <https://doi.org/10.1016/j.exmath.2010.09.005>
- [3] Walsh, J.B.: An introduction to stochastic partial differential equations. In: *École D’été de Probabilités de Saint-Flour, XIV—1984. Lecture Notes in Math.*, vol. 1180, pp. 265–439. Springer, Berlin (1986). <https://doi.org/10.1007/BFb0074920>
- [4] Da Prato, G., Zabczyzk, J.: *Stochastic Equations in Infinite Dimensions*, 2nd edn. *Encyclopedia of Mathematics and its Applications*, vol. 152, p. 493. Cambridge University Press, Cambridge (2014). <https://doi.org/10.1017/CB09781107295513>
- [5] Amir, G., Corwin, I., Quastel, J.: Probability distribution of the free energy of the continuum directed random polymer in $1 + 1$ dimensions. *Comm. Pure Appl. Math.* **64**(4), 466–537 (2011). <https://doi.org/10.1002/cpa.20347>
- [6] Misiats, O., Stanzhytskyi, O., Yip, N.K.: Existence and uniqueness of invariant measures for stochastic reaction-diffusion equations in unbounded domains. *J. Theoret. Probab.* **29**(3), 996–1026 (2016). <https://doi.org/10.1007/s10959-015-0606-z>
- [7] Misiats, O., Stanzhytskyi, O., Yip, N.K.: Invariant measures for stochastic reaction-diffusion equations with weakly dissipative nonlinearities. *Stochastics* **92**(8), 1197–1222 (2020). <https://doi.org/10.1080/17442508.2019.1691212>
- [8] Assing, S., Manthey, R.: Invariant measures for stochastic heat equations with unbounded coefficients. *Stochastic Process. Appl.* **103**(2), 237–256 (2003). [https://doi.org/10.1016/S0304-4149\(02\)00211-9](https://doi.org/10.1016/S0304-4149(02)00211-9)